



The Impact of Implementing Artificial Intelligence Technologies on improving Public Service Facilities in Algeria

Chibani Abdallah

Mohamed Lamine Debaghine setif 2 Universty, (Algeria)

a.chibani@univ-setif2.dz

Abstract

Algeria has witnessed a significant advancement in the digital transformation of its public administrations through the development of many digital applications and portals, easing access to require services and substantially enhancing the use of public services across various practical and scientific fields. Despite of aiming to improve the quality of public services and reinforce transparency and efficiency in Algerian public administrations, this digital transformation faces a lot of challenges, including inadequate technological infrastructure and limited public awareness of its importance.

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- ✓ *Administration*
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- ✓ *Artificial Intelligence.*

1. *Introduction*

In light of technological development, countries worldwide aim to leverage modern innovations, particularly Artificial Intelligence (AI) tools, to restructure public administrations seeking innovative mechanisms to facilitate and enhance the quality of public services, within the framework known as E. Government

New concepts have emerged around electronic or smart administration highlighting the role of intelligent technologies in improving and developing public facility management, accelerating the digitization of administrative procedures, and freeing them from traditional methods. Many projects have been launched to improve public service quality, boost efficiency, and ease interaction between citizens and government bodies.

This step is a key part of global efforts to promote intelligent technologies in enhancing administrative performance, contributing efficient resource use and better citizen services.

In this regard, Algeria is considered among the countries that have given considerable importance to the digital transformation of public administrations through innovative strategies that enable faster and more effective answering to citizen needs, thereby reinforcing satisfaction and trust in the state.

Importance of the research:

The importance of this topic is evident in several main points:

1- AI technologies can process and analyze vast amounts of data, extracting valuable and insights by examining citizen behaviors and preferences, helping understand their needs and meet various requirements.

2- AI improves public services by enhancing user experience and providing flexible, effective solutions to raise administrative process efficiency.

3- AI can predict potential threats and detect suspicious behavior patterns contributing in improving security and safety.

4- Generally, AI technologies are a fundamental means to modernize public facility management and make it more effective in line with continuous technological advancements.

Research objectives:

This research aims to know the importance of using AI tools and their impact on improving and enhancing the quality of public services.

Previous studies:

-Amina Hamdi, "Digitization of management as a mechanism to improve public service of facility", Emir

Abdelkader University magazine of Islamic sciences, Constantine, Vol.35 N° 2, 2021, concluded that E-administration is an effective mechanism for improving public administration performance by bringing administration closer to citizens, earning their satisfaction and trust, and addressing shortcomings of traditional administration.

-Rachid Sefahloo, "The role of public administration digitization in reducing financial Administrative corruption in Algeria (case of tax administration digitization), "Shoaa journal of economic studies, Tissemsilt University, vol 8, N°1, 2024.

found that E- administration help curb corruption in public institutions; fostering trust among citizens in dealing with administration.

Research Problem:

According to what have given above, the main research question is:

How do intelligent technologies contribute to up grading and improving public facility management?

This breaks down into:

- 1- What is meant by public facility in the context of intelligent technologies?
- 2- What are the most important AI applications in public facilities?.

To solve these problems, the research is divided into:

Chapter Ane: Basic Concepts

Section one: Public facilities in the context of modern technology.

Section two: Artificial Intelligence.

Chapter two: AI applications in Improving public facility services

Sections one: Using AI to Enhance Public Facility Performance.

Section two: Employing AI Technologies in Algerian Public facilities

Research Methodology:

The descriptive analytical method was used to describe and analyze the reality of public facilities under intelligent technologies field (Times New Roman13, interline 1.15)

2. Chapter One: Basic Concepts

Artificial intelligence applications currently cover all vital fields, including the field of administrative activity that significantly contribute to the development of public facilities in several fields. Before deliving into the concepts related to the topic of the study which includes public facilities (Sections one) and AI(Section two).

2.1 Subsection one: Electronic public facility

Before talking about the Electronic public facility, we must refer to its traditional-concept.

First: Traditional Public facility definition:

Traditionally, a public facility is considered as a positive representation of public administration activity, defined by organic and material theories Organic theory sees it as a set of state administrative bodies. Rather, the material one defines it as any activity conducted to achieve public interest¹.

However, relying on each criterion individually to define the concept of a public service was not sufficient. so, Modern doctrine combines both defining a public facility as any activity carried out directly or indirectly by public authority to serve the public interest².

Public Facility is considered one the most important elements of the basic infrastructure of the state, encompassing all activities must organize and oversee to meet individuals public needs and foster social solidarity, aiming to achieve social justice among citizens regardless of their differences in social, economic, or cultural backgrounds.

Second: Electronic Public Facility

The concept remains the same as traditional public facilities what changes are management means, now relying on modern technical and technological tools.

Electronic public facility is defined as any project managed by public authority directly or under its supervision to fulfill public interest and meet all individuals public needs using various modern technical technological means³.

In order to improve the role of public facilities, effective strategies must be adopted to organize and facilitate the facilities by enhancing the use of technology in the provision of public services, to improve their quality and increase their efficiency, so that they are always capable of meeting the various needs of society without obstacles, Thus, the public facility is a means aimed at facilitating the exchange of documents and administrative information between public and authorities, institutions, and individuals more quickly and easily.

Subsections two:

Legal Principles governing public facilities provide a variety of public services to meet the needs of society, they are governed by a set of legal principles that ensure the proper functioning of public services, aiming to improve the public facilities in a way that ensures their continuity and efficiency in serving the public interest.

First: Principles related to services beneficiaries⁴

1- The principle of equality, which means treating all individuals equally in benefiting from public services, without any discrimination based on gender, religion, race,

or others. This principle is considered a complement to the legal principle that all individuals are equal before the law.

2. The principle of continuity, which means that public facilities must remain active and continuously provide their services to meet the permanent and renewed needs of citizens. This principle to achieve it.

3- The principle of adaptability to changing needs, which means that public facilities must be able to keep up with social, economic, and technological developments to fulfill their mission efficiently.

Second: Principles related to public services:

1-The principle of continuity and regularity: Service should be delivered consistently through absolute continuity and regularity guaranteeing citizen's rights to get different services. though this principle can't be applied absolutely because public facilities are not obliged to work continuously and in all times so there is a relativity in its application, some conditions can be according to societies needs and basic public facilities⁵.

2- The principle of transparency and neutrality: services must be available to all without favoritism, enhancing trust and preventing administration corruption⁶.

Section two: Artificial Intelligence

Artificial Intelligence have an important role in developing public facilities in several

fields. In this section we will talk about AI concept (subsection one), and its application's importance in public facilities (subsection two).

Subsection one: Artificial Intelligence concept

Artificial Intelligence is considered as one of the important basics of new technologies, it is coined for its first time in 1956 to describe techniques which contain capacities and services related essentially to human intelligence, enabling problem-solving, reasoning, perceptions and communications.

AI systems can learn autonomously without explicit programming⁷, AI encompasses tools and electronic systems for generating, collecting, storing and processing data, aiming to perform tasks associated with human intelligence⁸ using machine learning, neural networks, advanced algorithms, and natural language processing to make autonomous decisions⁹.

Considering that technologies of AI develop quickly, it touches several fields of daily life, because it analyses huge quantities of data and it specifies patterns and behaviours and result predictions accurately. These are why we use to realize programs depending on AI in administrative services to enhance production efficiency.

2.2 Subsection two: Advantages of Ai in public facilities

AI applications in public facilities offer¹⁰:

- 1- New techniques offer high level of confidentiality and security for sensitive employee and client data.
- 2- biometric identification (eg: facial recognition) is considered one of the mains technics used in institutions, offering high level of security and accurating attendance tracking.
- 3- Intelligent staff selections and performance development through data analysis and trailored training recommendations.
- 4- Enhanced decision making quality by providing organized relevant information and automated insights.
- 5- Automation of financial processes and audits, ¹¹ increasing speed and precision compared to manual handling.
- 6- Combating administrative corruptions via transparent data tracking.

In short, artificial intelligence applications represent a great opportunity to develop public facilities and improve the quality of their services by enhancing efficiency and productivity especially as they continue to evolve -Undoubtedly, their adoption for the public good offers numerous benefits for institutions and governments on one hand, and for clients on the other.

3. Chapter Tow: AI Applications Improving Public Facility Services

Public needs have evolved making traditional methods insufficient necessitating technology driven approaches aligned with societal requirements.

Section one: Using Ai to enhance public Facility Performance, lestorting to technology, including artificial intelligence techniques is considered as an important step toward improving public facilities and ensuring their alignment with the changing needs of citizens through the transition to digital administration and the adoption of AI technologies to develop public services.

3.1 Subsection one: Digital Administrative Transformation in light of current changes, the labor market and management require reengineering and redesign of administrative processes and a shift toward smart management, also known as e- management, this involves the use of various technologies and information systems, which are considered among the most important tools contributing to the advancement performance. These technologies provide diverse and effective opportunities to enhance administrative processes such as plaining, organizing, and decision-making with accuracy and quickly. As a result, they enable the provision of distinguished and advanced services that keep pace with the evolving and ever-changing needs of society¹².

The process of digital or electronic transformation is not an easy or simple task. In fact, it requires careful evaluations of

various technologies and tools that are suitable for the work environment. A clear plan must be established with the aim of achieving specific goals, whether it is to improve efficiency, reduce waiting time, increase citizen satisfaction, or other objectives¹³.

Subsection two: Requirements for AI Adoption in developing public facility.

Developing public facilities and enhancing public services need working on many points to guarantee integration a successful AI integration.

1- Building infrastructure:

Integrating smart technologies into administrative work requires the development of a strategic infrastructure that includes both technical and legislative expertise. It is important to plan for transforming public facilities through smart technological approaches tailored to each sector, by relying on intelligent digital tools and methods that align with ongoing changes and the modernizations of public services¹⁴.

2- Developing a work Plan:

For the implementation of Artificial intelligence Systems and Smart Technologies is general, to ensure the success of any project, it is essential to have a clear and well-studied work plan that considers all aspects, especially regarding the feasibility of using artificial intelligence applications. This is particularly important in areas affected by

illiteracy and the digital divide regions where communication and internet services are either unavailable, weak or where is a lack of understanding of how to use such technologies. These challenges can render the implementations of smart technologies ineffective.

Therefore, it is necessary to search for practical solutions tailored to the needs and abilities of each region, by improving and strengthening the technological infrastructure and reducing the digital divide through raising community awareness and enhancing understanding of the importance of smart technologies in today's world¹⁵.

3- Eliminating Mental Rigidity Among Public Employees:

Modern technologies bring significant changes to work dynamics and the nature of jobs roles. While they enhance efficiency and productivity; they also impact how work is managed and executed. Although this transformation is not primarily about replacing humans with modern technologies, one of the biggest challenges to effective digital transformation is the fear of change and the concern among employees about losing their jobs.

Therefore, it is essential for employees to understand the role of smart technologies in developing and improving work processes. This can be achieved by training and upskilling staff in line with technological advancements, so they feel like part of the

solution rather than a replaceable component. The success digital transformation relies on several key factors, the most important of which are mature administrative thinking and flexibility in accepting various changes in the nature of work in all its forms¹⁶.

4-Using advanced geographic technology to determine the locations of public facilities:

Geographic technology is considered one of the key elements in accurately identifying optimal locations for public facilities. This plays a crucial role in enabling citizens to easily access services. Success in this field depends on the ability to collect, analyze, and process relevant data to reveal hidden relationships and insights that help is understanding situations and making appropriate decisions and solutions¹⁷.

5- Saving costs and time:

Many public service transactions share a similar structure, where clients are required to fill out forms, submit various supporting documents, and verify their identity. Some transactions also involve paying specific fees for some services. Therefore, integrating smart systems helps meet customer requirements and enhances flexibility and efficiency with less effort and cost. Smart public services can be accessible to the public and continue to operate under all conditions with establishing legal provisions, as was the case during the COVID-19 pandemic¹⁸.

6- Confidentiality and privacy:

Confidentiality and privacy are among the

most important pillars in serving of public services as they are crucial for building user trust. Therefore, the use of smart systems enhances the protection of personal data and prevents it from being breached, damaged altered, republished, or tampered with it from any one, ensuring high levels of user satisfaction¹⁹. By addressing these aspects through a comprehensive strategic approach, the desired goals of developing public can be achieved to better understand and meet the needs and expectations of citizens. This must be done while ensuring full compliance with cybersecurity requirements issued by the relevant authorities at all steps of designing, organizing, and improving public services.

Section two: Employing AI Technologies in Algerian Public Facilities

In an area dominated by technology, the search for innovative methods and mechanisms to make public services more efficient and higher quality has become a primary goal for all countries, including Algeria, which has made efforts to introduce several necessary improvements in this field. This section will address the legal basis for employing smart technologies in public facilities in Algeria (subsection one), as well as the manifestations of their implementation in these facilities (subsection two).

Subsection one: The legal basis for the use of smart technologies in public facilities.

The Algerian administration has undergone several reforms aimed at strengthening

administrative systems and processes by leveraging smart technologies to ensure the quality and efficiency of services and to enhance citizens satisfaction.

First: International Treaties:

Like other countries around the world, Algeria has shown interest in participating in conferences that promote the use of technology and artificial intelligence applications in the field of public services.

One of the most important of these was the²⁰ 2014 development and communication conference, which focused on enhancing and supporting social and economic development. Algeria participated in this conference, represented by the national telecommunications company- during the conference, Algeria acknowledged the significant role that information and communication technologies play in establishing advanced smart societies, as well as in promoting E. government activities for the benefit of developing countries by improving the quality of electronic public services and expanding the use of smart technologies across various aspects of daily life. This participation highlights the crucial importance of technology in strengthening and developing public facilities as part of notional strategies for the country's progress and advancement.

Second: The constitution

According to the Algerian constitution, public facilities hold great importance, with the

prime minister responsible for ensuring the good functioning of public services and public facilities²¹. This is achieved by focussing on improving the quality of services provided to citizens through the use of all available means, including smart technologies. In this context, Algeria has restructured its traditional administration and shifted toward e-governance. This transition is expected to offer many effective solutions for developing and enhancing public services. Technology and smart systems enable the use of data in a more effective and transparent manner by processing and analyzing it through advanced methods, thereby improving the efficiency and quality of services delivered to citizens. Ultimately, this contributes to a better quality of life and fosters comprehensive development across the country.

3.2 Subsection two: Manifestations of the use of Smart Technologies in Public Facilities

The steps of integrating smart technologies into Algerian public facilities are highlighted in:

First: The E-Government Project

This projet is considered one of the most important initiatives adopted since 2008 by the ministry of post, information, and communication technologies in Algeria (TIC). It aims to accelerate the use of information and communication technologies within public administration to ensure the delivery of

more effective and transparent public services to citizens. In this context, 13 Key ascis were identified to address various sectors, including accelerating the adoption of technology as much as possible, and strengthening communication between citizens and public administration.

The initiative also focuses on enhancing technological infrastructure and developing electronic public services²².

Despite the importance of the project, its full implementation faced numerous challenges. among the most prominent were lack of awareness about significance of e-government and its role in the most and accelerating administrative processes and procedures; widespread digital illiteracy among both employees and citizens, a shortage of specialized technical expertise in this field and-concerns over potential breaches of citizens' personal data. Additionally, there were legislative limitations regarding electronic transactions, with further hindered progress²³.

To overcome these obstacles, it is essential to continue investing in the development and improvement of technological infrastructure, strengthen support for training programs and the development of both government employees and citizens, and promote digital culture. These efforts aim to accelerate and expand the use of smart technologies, thereby enhancing the efficiency of public administrations and improving the quality

electronic public services, ensuring that everyone benefits from their advantages.

Second: The National Observatory of Public Service

The National Observatory of Public Service is one of the advisory bodies of the state. It was established by presidential decree N: 16-03²⁴.

Its primary mission is to promote and modernize public service in response to current technological changes and challenges. It strives to:²⁵

1- Enhance the status and role of the citizen:

The National Observatory works to improve the relationship between the administration and the citizen by building mutual trust through enhancing the quality of public services provided to citizens and addressing various shortcomings and issues.

2- The contribution in the modernization of public facilities in line with citizens' needs and technological developments across all sectors, the Notional Observatory studies the conditions of public administrations and administrative procedures, providing necessary recommendations for their modernization.

3- Electronic administrations development:

The Notional Observatory seeks to promote the widespread use of technology across all sectors by adopting electronic procedures in various fields, ensuring that citizens can access information of and services easily and

efficiently.

4- Improving communication and transparency:

The Notional of Observatory works on establishing effective systems and procedures for communication with the aim of informing citizens about the available public services, improving administrative processes, and answering to their complains, opinions, and suggestions in order to find appropriate solutions to existing shortcomings.

5- providing consultations:

As the National Observatory is an advisory body, it can provide all relevant suggestions and solutions it deems appropriate for developing public facilities and improving the quality of their services.

However, it should be noted that these suggestions are not legally binding, and the decision to adopt them or not remains within the authority of the competent bodies²⁶.

Third: The National School of Artificial intelligence:

This school was inaugurated in the 2011/2012 academic year and is considered a significant turning point in the educational and

technological landscape of the country. It aims to foster innovation to drive the advancement of scientific research and to enhance the integration of artificial intelligence technologies into public facilities. The school is specifically designed to prepare, train, and educate student in data science and artificial intelligence, equipping them to face future challenges in fields such as energy, healthcare, agriculture and transportation among others. These fields, in turn, contribute to the continuous improvement and modernization of services provided to citizens, in line with the evolving changes in society²⁷.

These applications work to implement and enforce national policies related to public facilities and aim to address various issues related to administration processes. They are considered a valuable asset in promoting public services in the country by improving service delivery and meeting public needs through identifying shortcomings and imbalances and seeking appropriate solutions. these initiatives come within the context of technological development and the enhancement of the relationship between citizens and public administration.

achieving a balance between the various dimensions of development and ensuring social solidarity. Undoubtedly, achieving these goals requires continuous efforts and strong cooperation among all concerned parties, starting from the government down to the ordinary citizen. This can be accomplished through the adoption of

4. CONCLUSION

At the end of this study, it must be acknowledged that the use artificial intelligence applications in the development of public spaces play a crucial role in enhancing the efficiency and quality of services provided to citizens. It contributes to

comprehensive and effective policies aimed at improving the performance of public services and regulating their operations efficiently.

Recommendations:

1. The necessity of developing electronic infrastructure to meet citizens' public needs more quickly and efficiently.
2. Eliminate digital illiteracy, starting from administrative institutions to ordinary citizens, by promoting a culture of digital transformation. This includes opportunities for continuous learning training and development, as well as encouraging creativity, innovation, and research in this field.
3. Improving the relationship between public administration and citizens and encouraging digital communication between them to better understand their needs and desires, in order to develop public facilities and enhance the quality of services in a way that better meets the evolving and renewed demands of society

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